Equation

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$$
\begin{aligned}
\text{OGseed_percent_maincrop}_i &\sim N\left(\mu, \sigma^2\right) \\
\mu &= \alpha_{j[i],k[i]} + \beta_1(\text{legitimation_user}) + \beta_2(\text{CertifierRequestn}) + \beta_3(\text{noncommercial_seedsource_s} \\
\alpha_j &\sim N\left(\gamma_0^\alpha + \gamma_1^\alpha(\text{entrants_supplychain_n_avg_s}) + \gamma_2^\alpha(\text{resources_funding_100thous}) + \gamma_3^\alpha(\text{kno}\right. \\
\alpha_k &\sim N\left(\mu_{\alpha_k}, \sigma^2_{\alpha_k}\right), \text{ for maincrop k} = 1, \ldots, \text{K}
\end{aligned}
$$

$$
\begin{aligned}
\text{Organic seed acreage (percent)}_i &\sim N\left(\mu, \sigma^2\right) \\
\mu &= \alpha_{j[i],k[i]} + \\
&\quad \beta_1(\text{Farm size}) + \\
&\quad \beta_2(\text{Non commercial seed sourcing}) + \\
&\quad \beta_3(\text{Value organic seed}) + \\
&\quad \beta_4(\text{Certifier request}) + \\
\alpha_j &\sim N(\gamma_0^\alpha + \\
&\quad \gamma_1^\alpha(\text{Supplier entrants}) + \\
&\quad \gamma_2^\alpha(\text{Research funding}) + \\
&\quad \gamma_3^\alpha(\text{Organic IP}), \\
&\quad \sigma^2_{\alpha_j}), \text{ for Region j} = 1, \ldots, \text{J} \\
\alpha_k &\sim N\left(\mu_{\alpha_k}, \sigma^2_{\alpha_k}\right), \text{ for Main crop category k} = 1, \ldots, \text{K}
\end{aligned}
$$